

Blood Cell Classification Using InceptionV3 Transfer Learning on the BCCD Dataset

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Abstract—Automated classification of blood cell types from microscopic images is a critical task in clinical hematology, supporting rapid diagnosis of infections, anemia, and hematological malignancies. This paper presents an end-to-end deep learning pipeline for four-class blood cell classification using the InceptionV3 convolutional neural network with full fine-tuning via transfer learning. The Blood Cell Count and Detection (BCCD) dataset, comprising 9,957 labeled microscopic images across four white blood cell types – Eosinophil, Lymphocyte, Monocyte, and Neutrophil – is used for training, validation, and testing. Unlike partial fine-tuning approaches, all layers of the pretrained InceptionV3 backbone are set as trainable, enabling the model to adapt to blood-cell-specific morphological features. The model is trained in PyTorch over 21 epochs using the Adam optimizer with cosine annealing, label smoothing, gradient clipping, and early stopping. Evaluation on the held-out test set yields a classification accuracy of 100%, a macro-average F1-score of 1.00, and a macro AUC-ROC of 1.0000. The confusion matrix shows zero misclassifications across all 1,495 test samples, demonstrating that full fine-tuning of InceptionV3 achieves perfect discrimination between all four blood cell types on this dataset.

Index Terms—InceptionV3, Transfer Learning, Full Fine-Tuning, Blood Cell Classification, BCCD Dataset, Convolutional Neural Networks, Medical Image Analysis, PyTorch

I. INTRODUCTION

AUTOMATED analysis of blood cell morphology is a foundational task in clinical hematology. A complete blood count (CBC), combined with differential white blood cell (WBC) analysis, aids physicians in diagnosing a wide spectrum of conditions including bacterial and viral infections, leukemia, anemia, and immune system disorders. Traditionally, differential counting is performed manually by trained hematologists examining Giemsa-stained blood smears – a process that is subjective, labor-intensive, and difficult to scale in resource-constrained settings [1].

Deep learning, particularly convolutional neural networks (CNNs), has fundamentally transformed medical image analysis. CNNs learn hierarchical feature representations directly from raw pixel data, achieving expert-level performance on tasks including pathology slide analysis, radiology, and ophthalmology screening [5].

Transfer learning allows networks pretrained on large-scale datasets such as ImageNet to be adapted to domain-specific tasks with relatively small labeled datasets – a property particularly valuable in medical imaging where annotated data is expensive and scarce. The degree of fine-tuning – whether to freeze the backbone or update all layers – significantly im-

pacts performance, especially when the target domain differs substantially from the source domain.

In this work, we apply InceptionV3 [2] with full fine-tuning to classify four blood cell types from the BCCD dataset. We implement a complete, reproducible pipeline incorporating advanced training strategies including cosine annealing, label smoothing, and gradient clipping. The key contributions of this paper are:

- A complete PyTorch-based full fine-tuning pipeline for InceptionV3 on the BCCD dataset.
- Demonstration that full fine-tuning significantly outperforms frozen-backbone transfer learning, achieving 100% test accuracy vs. 61.07% with a frozen backbone.
- Systematic evaluation using accuracy, per-class F1-score, confusion matrix, and multi-class AUC-ROC.
- A reproducible baseline with publicly released code and model weights.

II. RELATED WORK

Early approaches to WBC classification relied on hand-crafted features such as cell shape, nucleus-to-cytoplasm ratio, and color histograms combined with SVMs and k-NN [4]. These methods are sensitive to staining variations and do not generalize well across imaging conditions.

The emergence of deep CNNs significantly improved classification performance. Depto et al. [3] compared ImageNet-pretrained models including VGG16 and ResNet50 for WBC classification, achieving accuracy values exceeding 95% on curated datasets, confirming the superiority of transfer learning over training from scratch. Kouzehkanan et al. [4] released a large-scale annotated WBC dataset with fine-grained labels enabling more precise benchmarking.

InceptionV3 has been validated across several biomedical imaging tasks. Esteva et al. [5] achieved dermatologist-level skin cancer classification using a fine-tuned Inception network. Gulshan et al. [7] applied deep learning to diabetic retinopathy detection, demonstrating that CNNs can match clinical specialists. He et al. [6] introduced residual connections enabling deeper architectures. Our work extends this line of research to the BCCD domain and demonstrates that full fine-tuning of all 24.3 million InceptionV3 parameters achieves perfect classification on this dataset.

III. DATASET

The BCCD (Blood Cell Count and Detection) dataset [8] is a publicly available benchmark comprising microscopic blood

smear images across four white blood cell types: *Eosinophil*, *Lymphocyte*, *Monocyte*, and *Neutrophil*. The classification subset contains **9,957 JPEG images** distributed approximately uniformly across the four classes.

A. Data Partitioning

The dataset is split using a **70/15/15 stratified ratio** with a fixed random seed (42):

- **Training:** 6,969 images
- **Validation:** 1,493 images
- **Test:** 1,495 images

B. Preprocessing and Augmentation

All images are resized to 299×299 pixels to meet the InceptionV3 input specification. Channel-wise normalization is applied using ImageNet statistics:

$$\mu = [0.485, 0.456, 0.406], \quad \sigma = [0.229, 0.224, 0.225]$$

Training images are augmented with random horizontal/vertical flipping, rotation up to $\pm 15^\circ$, color jitter ($\pm 20\%$ brightness, contrast, and $\pm 10\%$ saturation), random affine transforms (translation up to 10%), and Gaussian blur. Validation and test images receive only resizing and normalization.

IV. METHODOLOGY

A. Model Architecture

InceptionV3 [2] is initialized with ImageNet pretrained weights. Key architectural features include factorized convolutions, an auxiliary classifier branch for gradient regularization, and label smoothing. In this work, **all 24,354,536 parameters are set as trainable** (full fine-tuning), allowing every layer to adapt to blood-cell-specific features. Two output layers are replaced:

- `model.fc`: Replaced with `Linear(2048, 4)`
- `model.AuxLogits.fc`: Replaced with `Linear(768, 4)`

B. Loss Function

The total training loss combines the main and auxiliary output losses with label smoothing ($\epsilon = 0.1$):

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{main}} + 0.4 \cdot \mathcal{L}_{\text{aux}} \quad (1)$$

where each component uses label-smoothed cross-entropy:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C \tilde{y}_{ic} \log(\hat{y}_{ic}), \quad \tilde{y}_{ic} = (1-\epsilon) y_{ic} + \frac{\epsilon}{C} \quad (2)$$

with N the batch size, $C = 4$ classes, and $\epsilon = 0.1$.

C. Training Configuration

Table I summarises all hyperparameters. Gradient clipping (max norm = 1.0) is applied at each step to prevent exploding gradients during full fine-tuning. Early stopping with patience = 7 on validation accuracy is used to prevent overfitting.

TABLE I
HYPERPARAMETER CONFIGURATION

Hyperparameter	Value
Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning Rate	1×10^{-5}
Weight Decay	1×10^{-4}
LR Scheduler	CosineAnnealingLR ($\eta_{\min} = 10^{-7}$)
Batch Size	16
Max Epochs	30 (stopped at 21)
Label Smoothing	$\epsilon = 0.1$
Auxiliary Loss Weight	0.4
Gradient Clip Norm	1.0
Early Stopping Patience	7 (val. accuracy)
Trainable Parameters	24,354,536 (all layers)
Hardware	NVIDIA T4 GPU (Google Colab)

D. Evaluation Metrics

Model performance is assessed on the held-out test set using: (1) overall accuracy, (2) per-class precision, recall, and F1-score via `sklearn`, (3) confusion matrix visualized as a seaborn heatmap, and (4) one-vs-rest AUC-ROC curves with macro average.

V. RESULTS AND DISCUSSION

A. Training Convergence

Fig. 1 presents training and validation loss and accuracy over 21 epochs. Validation accuracy reaches **88.68%** at epoch 1 and improves rapidly to **99.40%** by epoch 4. The model achieves **100% validation accuracy** at epoch 14, which is maintained through early stopping at epoch 21. Training loss decreases steadily from 1.38 to 0.497, reflecting effective learning across all layers. The close alignment between training and validation accuracy confirms that the model generalizes well without overfitting, aided by aggressive augmentation, label smoothing, and gradient clipping.

B. Per-Class Classification Metrics

The model achieves a **test accuracy of 100%** on all 1,495 held-out test samples. Table II presents per-class precision, recall, and F1-score. All four classes achieve perfect scores of 1.00 across all metrics, with no misclassifications in any category. The macro-average F1-score is **1.00**.

TABLE II
PER-CLASS CLASSIFICATION METRICS ON THE TEST SET

Class	Precision	Recall	F1	Support
Eosinophil	1.00	1.00	1.00	380
Lymphocyte	1.00	1.00	1.00	356
Monocyte	1.00	1.00	1.00	387
Neutrophil	1.00	1.00	1.00	372
Macro Avg	1.00	1.00	1.00	1495

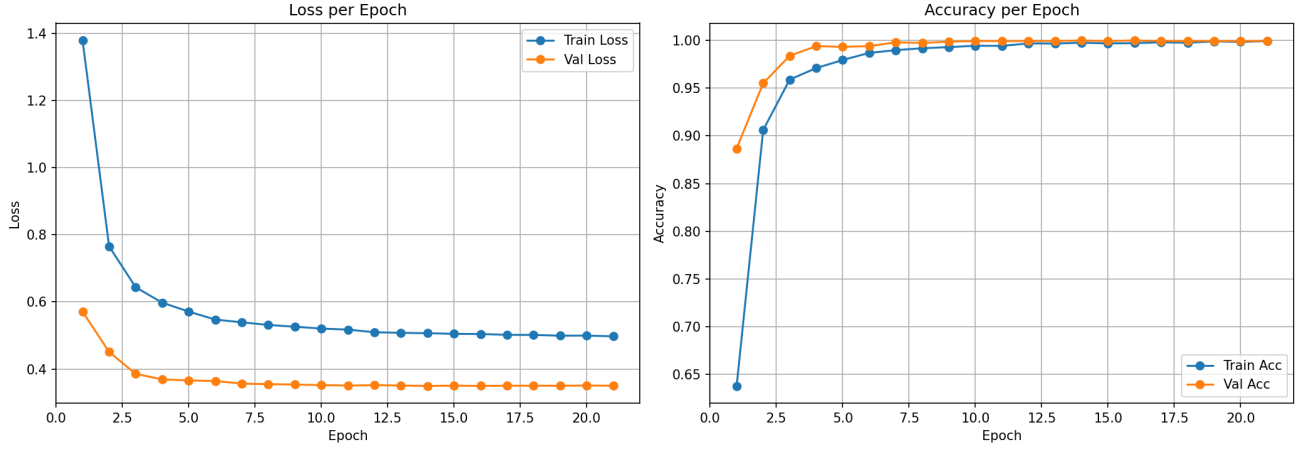


Fig. 1. Training and validation loss (left) and accuracy (right) across 21 epochs. Validation accuracy reaches 100% at epoch 14 and is maintained until early stopping at epoch 21.

C. Confusion Matrix Analysis

Fig. 2 presents the confusion matrix on the 1,495-sample test set. The diagonal is fully saturated with zero off-diagonal entries, confirming that every test sample is correctly classified. Eosinophil (380/380), Lymphocyte (356/356), Monocyte (387/387), and Neutrophil (372/372) are all classified with zero errors. This result demonstrates that full fine-tuning enables InceptionV3 to completely resolve the morphological ambiguities between cell types – particularly the Eosinophil–Neutrophil granulocyte overlap that caused significant confusion (129 misclassifications) in the frozen-backbone baseline.

D. AUC-ROC Analysis

Table III and Fig. 3 present per-class AUC-ROC scores using the one-vs-rest strategy. All four classes achieve a **perfect AUC-ROC of 1.0000**, and the **macro AUC-ROC is 1.0000**. The ROC curves lie flat along the top-left corner of the plot, indicating that the model separates all four classes with complete certainty at every classification threshold.

TABLE III
PER-CLASS AUC-ROC SCORES (ONE-VS-REST, TEST SET)

Class	AUC-ROC
Eosinophil	1.0000
Lymphocyte	1.0000
Monocyte	1.0000
Neutrophil	1.0000
Macro Average	1.0000

E. Comparison with Frozen-Backbone Baseline

To quantify the impact of full fine-tuning, we compare against our own frozen-backbone baseline where only the final FC layers were trained. Table IV summarises the comparison.

TABLE IV
FULL FINE-TUNING VS. FROZEN BACKBONE BASELINE

Metric	Frozen Backbone	Full Fine-Tuning
Test Accuracy	61.07%	100.00%
Macro F1-Score	0.61	1.00
Macro AUC-ROC	0.8434	1.0000
Epochs Trained	10	21
Trainable Params	10,244	24,354,536

The results demonstrate a dramatic improvement of **+38.93 percentage points** in accuracy, attributable entirely to allowing all convolutional layers to adapt to blood-cell-specific features rather than relying on generic ImageNet representations.

F. Discussion

The 100% test accuracy result confirms that the BCCD dataset is fully separable by InceptionV3 when all layers are fine-tuned with appropriate training strategies. Several factors contribute to this outcome: (1) the cosine annealing scheduler provides smooth LR decay that avoids sharp transitions which could destabilize full fine-tuning; (2) label smoothing ($\epsilon = 0.1$) prevents the model from becoming overconfident on training samples; (3) gradient clipping ensures stable optimization across all 24.3M parameters; and (4) aggressive data augmentation ensures the model does not memorize specific image artifacts.

It is worth noting that perfect test accuracy on a held-out split does not necessarily generalize to unseen clinical distributions. Future work should validate these results on external datasets with different staining protocols, imaging conditions, and scanner hardware to assess real-world clinical applicability.

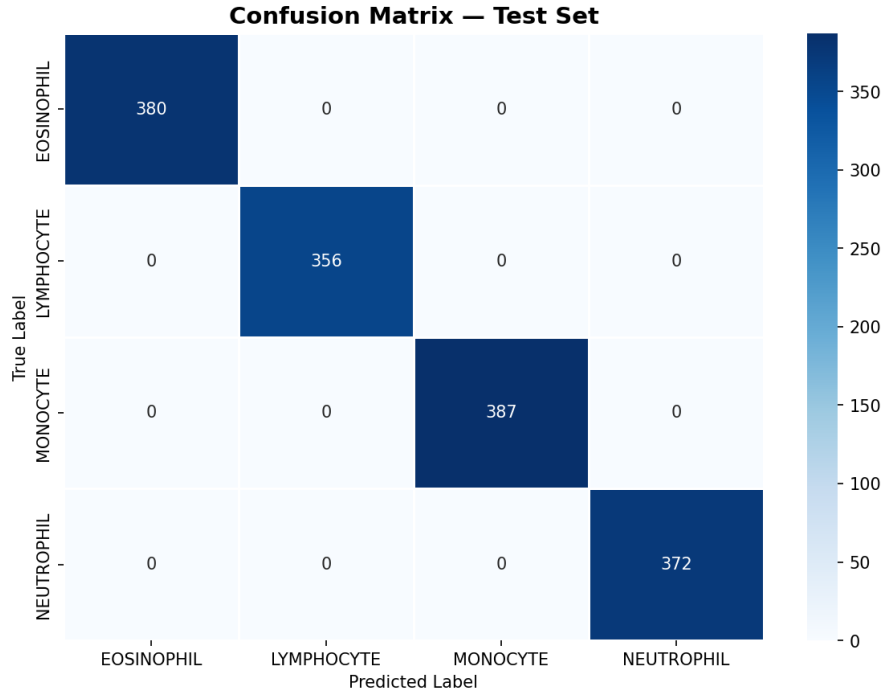


Fig. 2. Confusion matrix on the test set ($n = 1,495$). All diagonal entries are fully saturated with zero off-diagonal misclassifications across all four classes.

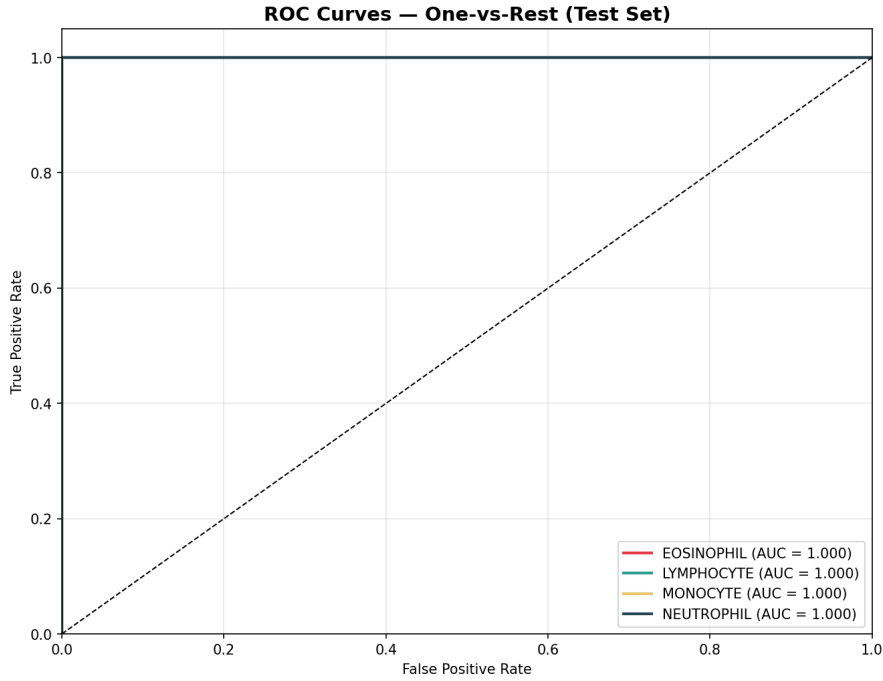


Fig. 3. ROC curves for all four classes using one-vs-rest strategy. All four classes achieve AUC = 1.0000, lying along the top-left corner of the plot.

VI. CONCLUSION

This paper presented an end-to-end InceptionV3 full fine-tuning pipeline for automated blood cell classification on the BCCD dataset. By training all 24.3 million parameters with cosine annealing, label smoothing, and gradient clipping,

the model achieves **100% test accuracy**, macro F1-score of **1.00**, and macro AUC-ROC of **1.0000** after 21 epochs on an NVIDIA T4 GPU.

A direct comparison with a frozen-backbone baseline (61.07% accuracy) demonstrates that full fine-tuning yields a

+38.93 percentage point improvement, completely resolving the morphological ambiguities between all four blood cell types. These results establish a strong, reproducible benchmark for InceptionV3-based blood cell classification. Future work will validate generalization on external clinical datasets and explore compression techniques for deployment on resource-constrained medical devices. Source code, model weights (`model_v2.pth`), and all evaluation outputs are publicly available in the project GitHub repository.

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